

Arsitektur BDAT — Simaster UGM vNEXT (3 Aplikasi: Staff, Student, Parent)

Tipe dokumen: Arsitektur aplikasi eksisting (reverse engineering) **Sumber:** APK Staff 1.8.7 + Student 1.9.6 + Parent 1.3.8 (strings + blutter Dart AOT) **Status:** Lengkap utk layer Application/Technology/Data; Business = inferensi dari modul **Repo:** simaster-vnext-re

1. BUSINESS LAYER

Visi Layanan

Ekosistem mobile UGM (3 peran): Staff/dosen (produktivitas kerja), Mahasiswa (akademik + transportasi), Orang Tua (pantau anak).

Proses Bisnis per Aplikasi

Proses	Staff 1.8.7	Student 1.9.6	Parent 1.3.8
Presensi kehadiran	attedence, attendance_gps, checkpoint	attedence, personal-qr, qr-scanner	attedence
Kinerja/IKU	IKU, activity	—	—
Bimbingan akademik	bimbingan	bimbingan	—
Administrasi surat & TTE	letters, electronic_signature, dms, tte	—	—
Akademik (jadwal/nilai)	—	course-schedule, exam-schedule, study-result, tugas-akhir	course-schedule, exam-schedule
TransGama (bus realtime)	transgama	transgama + armada GPS (haltes, putaran, jam)	—
Kendaraan/gate	vehicles, my_gate_vehicle	my-gate, my-gate-vehicle	—
Klaim anak (parenting)	—	—	claim_child, claim_parent
Dashboard pantau anak	—	—	child-dashboard, children
Beasiswa	—	scholarship, history-scholarship	scholarship
Pembayaran	—	—	history/payments
Kesehatan/klinik	clinic info	home-information-detail-service-clinic	home-information-detail-service-clinic
Laporan/OSS	oss	oss_v2, lapor (laporkan.ugm.ac.id)	oss
Auth	UGM/Google/Facebook	UGM/Google/Facebook	UGM/Google/Facebook/ Microsoft

Aktor

Dosen/Tendik (Staff) · Mahasiswa (Student) · Orang Tua (Parent) · Admin

2. DATA LAYER

Domain	Staff	Student	Parent
Identitas	profil (NIK, unit)	profil	profil orang tua
Presensi	log, rekap	personal QR	—
Akademik	kinerja, publikasi	KRS, jadwal, nilai, tugas akhir	jadwal anak
Transportasi	—	armada GPS, halte, putaran	—
Kuangan	—	—	payment history
Relasi keluarga	—	—	anak-anak (claim)
Konten	banners	banners, berita	banners

3. APPLICATION LAYER

Komponen	Teknologi	Bukti
Framework	Flutter (Dart 3.8.0 AOT)	libapp.so 15-19MB, blutter reconstruct
Package	simaster_ugm + simaster_core	blutter pp.txt
State/DI	GetX pattern	controllers/bindings/views
Network	Dio (HttpClient)	Bearer + BaseOptions
Auth	OAuth2 authorization_code + PKCE S256	OIDC discovery, client per-role
Push	FCM	firebase_api module
Map	OSM	oss/oss_v2
QR	qr-scanner, personal-qr	attedence

Client OAuth (verified)

App	Client ID	Status
Staff	7b7a2183da054558b172db0e667586d9dQ9Q	prod, public+PKCE
Student	e6abd4e380a5462e83873fe22ab8c219yVaU	prod, public+PKCE
Parent	(belum ditemukan — strings hanya SHA-1 hash)	perlu blutter dalam

4. TECHNOLOGY LAYER

Komponen	Detail
API gateway	api.simaster.ugm.ac.id/vnext/v1/
OAuth	oauth.simaster.ugm.ac.id (prod) + oauth.dev.ugm.ac.id (dev — bocor di prod APK)
SSO	sso.ugm.ac.id CAS (federated via Google)
File	files.simaster.ugm.ac.id/get/file.php
GPS armada	armada.smartcampus.ugm.ac.id/gps (TransGama)
Laporan	laporkan.ugm.ac.id
Kesehatan	api.checking.ugm.ac.id (RSA UGM)
Endpoint total	~23 API + 90+ routes lintas 3 app

5. TEMUAN KEAMANAN (konsolidasi)

1. **CRITICAL** — **oauth.pstmn.io/v1/browser-callback** di Student production = dev OAuth client (Postman) aktif di prod
2. **oauth.dev.ugm.ac.id LIVE** di ketiga APK production — endpoint dev (authorize/token) masih aktif
3. **armada.smartcampus.ugm.ac.id/gps** — GPS realtime kendaraan TransGama (risiko tracking tanpa auth kuat)
4. **Pola client ID**: 32hex+alnum = client valid; 40hex = SHA-1 (bukan client)
5. Dev OAuth client di APK prod (staff & parent) — authorize google/facebook/microsoft di dev

6. KARAKTER ARSITEKTUR

- **Ekosistem multi-role**: 3 APK terpisah (peran beda) berbagi gateway `api.simaster.ugm.ac.id/vnext/v1/`
- **Client per-role**: masing-masing app punya client OAuth sendiri (staff ≠ student)
- **Flutter monolith per app** — kode Dart AOT, endpoint dinamis (perlu blutter)
- **Keamanan berlapis**: OAuth2 + PKCE + SSO CAS + 2FA Google

7. DOKUMEN TERKAIT

- LAPORAN-ENDPOINT-20260812.md (staff 39 endpoint)
- LAPORAN-BLUTTER-20260812.md (staff blutter)
- LAPORAN-ENDPOINT-STUDENT-PARENT-20260813.md (student 10 + parent 13 endpoint)
- LAPORAN-OAUTH-STUDENT-PARENT-20260813.md (client ID + grant)
- endpoints.json / endpoints-student.json / endpoints-parent.json